

Area Surface Area Volume Practice

mathNexa

Name:

Date:

Success Criteria:

- Calculate areas of triangles, parallelograms, and polygons.
- Determine surface areas using nets and prisms.
- Find volumes of rectangular prisms with fractions.

1. A rectangular community garden plot measures 12 m long and 8.5 m wide. What is the total area of the garden plot?



A 41 m²

B 96.5 m²

C 102 m²

D 102 m

2. A rectangular wooden storage crate has a length of 6 ft , a width of 4 ft , and a height of 5 ft . What is the volume of the storage crate?



A 15 ft³

B 74 ft²

C 60 ft³

D 120 ft³

3. A triangular fabric sail has a base of 7 m and a perpendicular height of 10 m . What is the area of the sail?



- A 70 m² B 17 m² C 35 m D 35 m²
4. The net of a rectangular cereal box unfolds into 6 rectangular faces with dimensions: two 8 cm by 5 cm faces, two 8 cm by 20 cm faces, and two 5 cm by 20 cm faces. What is the total surface area of the box?



- A 800 cm³ B 600 cm²
C 520 cm² D 300 cm²
5. A glass terrarium is packed completely with $\frac{1}{2}$ -inch unit cubes. If the terrarium measures $4\frac{1}{2}$ in long, 3 in wide, and $2\frac{1}{2}$ in high, how many unit cubes fit inside, and what is its total volume in cubic inches?



- A 10 cubes, 33.75 in³ B 270 cubes, 33.75 in³
C 135 cubes, 33.75 in³ D 270 cubes, 67.5 in³

6. A stone paving slab in a courtyard is shaped like a parallelogram with a base of 15 in and a perpendicular height of 9 in. What is the area of the slab?



Answer: _____ sq in

7. A wooden jewelry box shaped like a rectangular prism has length 10 cm, width 6 cm, and height 4 cm. What is the total surface area of the box?



Answer: _____ sq cm

8. A raised garden planter bed has a length of $3\frac{1}{2}$ ft, a width of 2 ft, and a depth of $1\frac{1}{2}$ ft. What volume of potting soil is needed to fill the planter completely?



Answer: _____ cu ft

9. An L-shaped bedroom floor is formed by two adjoining rectangles. One section measures 14 ft by 8 ft, and the connecting section measures 6 ft by 5 ft. What is the total floor area to be carpeted?



Answer: _____ sq ft

10. A camping tent is shaped like a right triangular prism. The two triangular ends each have a base of 6 ft and a height of 4 ft (with side edges of 5 ft). The length of the tent is 8 ft, and the rectangular floor is 6 ft by 8 ft. What is the total surface area of the tent (including the floor)?



Answer: _____ sq ft

Answers

1. C

2. D

3. D

4. B

5. B

6. 135

7. 248

8. 10.5

9. 142

10. 152